

FEDERAL POLYTECHNIC MUBI, ADAMAWA STATE
COMPUTER SCIENCE DEPARTMENT

LECTURE NOTE

ON

COM III

INTRODUCTION TO COMPUTER

1. WHAT IS A COMPUTER?

The term computer is derived from the word *compute*. The word *compute* means to *calculate*.

A *computer* is an electronic machine that accepts data from the user, processes the data by performing calculations and operations on it, and generates the desired output results.

Computer performs both simple and complex operations, with speed and accuracy.

Though a computer is a machine it produces 100% correct result. It does never a mistake, because it produces desired information according to our input data. If we programmed properly it will come up with the most accurate answer. That is why the term GIGO (Garbage In Garbage Out) is associated with accuracy of computer.



Data: It is the collection of unstructured facts, figures & symbols that have not been processed by humans or with a computer, from which information and knowledge are derived.

Example: Names of students and their marks in different subjects listed in random order.

Information: It is the data that is processed and presented in an organized manner.

Example: When the names of students are arranged in alphabetical order, total and average marks are calculated & presented in a tabular form, it is information.

Program: Set of instructions that enable a computer to perform a given task.

Processing: is the method of converting the input data into the required output form with the help of the computer program and hardware.

Functionalities of a Computer

Any digital computer works as *an Input-Process-output system* and carries out the following functions/operations in sequence:

1. Input: Takes data as input and command from the input device to the computer.
2. Processing: it Processes the data and converts it into a useful format.
3. Output: it generates the output which is displayed by the computer to the user, either hard copy or soft copy.
4. Storage: it Stores the data/instructions in its memory and uses them as required inside or outside the computer.
5. Control all the above four steps.

Advantages of Computers

The following are certain advantages of computers.

1. **High Speed:** The computer is a very fast device that can perform millions of calculations in a few seconds (in a microsecond, nanosecond, and even the picosecond) as compared to a man.
2. **Accuracy:** Computers perform thousands of operations with 100% accuracy provided that the input is correct.
3. **Storage Capability:** A computer has much more storage capacity than human beings that can store a large amount of data in small space such as images, videos, text, audio, etc.
4. **Diligence:** Unlike human beings, a computer can perform jobs for a long time obediently, free from monotony, tiredness, and lack of concentration.
5. **Versatility:** It can be used to solve multiple (tedious) tasks in various domains at the same time.
6. **Reliability:** A computer does exactly what it is told to do without human intervention.
7. **Automation:** A computer is an automatic machine that can perform a given task automatically under the control of a stored program.
8. **Communication:** A computer can connect to *share resources*.
9. **Power:** the larger the word length of a computer the more powerful it becomes
10. **Paperless processing:** data can be processed using computers instead of papers

Disadvantages of Computers

Though computers are intelligent machines:

- i. No I.Q. A computer cannot take any decision on its own.
- ii. It cannot correct any wrong instruction fed in by an operator, it is Garbage-In-Garbage-Out
- iii. Dependency: It functions unless programmed; thus, it is fully dependent on humans.
- iv. A computer cannot decide how they are to be programmed or provide input.
- v. A computer cannot implement their decisions or interpret their results. A child can easily tell the difference between a cat and a sheep while a computer cannot.
- vi. No Feeling: Computers have no feelings or emotions and so cannot make judgments based on feeling, taste, experience, and knowledge, unlike humans.

4. COMPUTER HARDWARE COMPONENTS

A computer is made up of two constituents: hardware and software.

The term *hardware* refers to all of the physical devices, or *components*, that a computer is made of. A computer is not one single device, but a system of devices that all work together. A typical computer system consists of five major categories of hardware components (devices): Processing devices, Secondary storage devices, Input devices, main memory, and Output devices

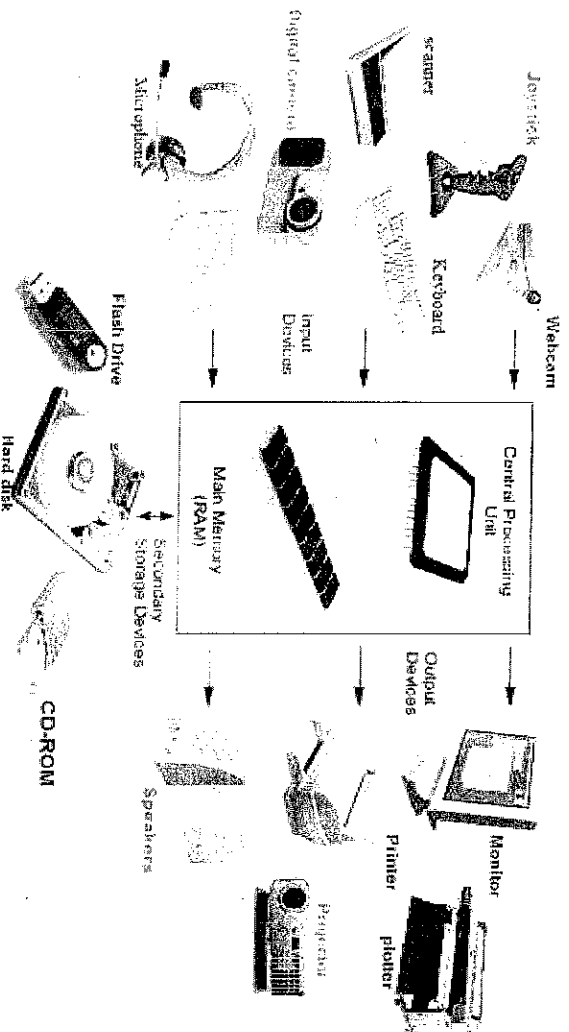


Fig.2: Computer hardware's

The system unit is the core of a computer system. Usually, it's a rectangular box that houses many electronic components that process information. The most important of these components is the central processing unit (CPU), or microprocessor, which acts as the "brain" of your computer. **Processing devices:** These are the devices in the processing unit to interpret instructions, process data, and control the other devices. They include CPU, ALU, control unit, and main memory.

The CPU: The *central processing unit*, or *CPU*, is the part of a computer that runs programs. The CPU consists of the Arithmetic Logic Unit (ALU) and the control unit.



Fig.3: CPU

ALU: It is where all arithmetic operations and logical functions are performed.

The Main Memory / Primary Memory Units: It stores data, program instructions, internal results, and final output temporarily before it is sent to an appropriate output device. They are:

- ✓ RAM (Random Access Memory) - Active, temporary.
- ✓ ROM (Read-only Memory) - Permanent

RAM is a temporary storage device or primary workspace that stores information that the CPU uses

while the computer is on. RAM is a memory that does not hold its contents without power. ROM is a special type of memory that can only be read and contents of which are not lost even when the computer is switched off. It typically contains the manufacturer's instructions and 'bootstrap loader' whose function is to start the computer software operating, once the power is turned on.

Input Devices: The component that collects the data and sends it to the computer is called an *input device*. They allow you to enter data, programs, commands, and user responses into a computer. Common input devices are the keyboard, mouse, scanner, microphone, and digital camera. Disk drives and optical drives can also be considered input devices because programs and data are retrieved from them and loaded into the computer's memory.

Keyboard: This is a peripheral device used to enter data, feed information to the system, or to communicate with the system. When you press a key on the keyboard, the action sends a signal to the system processing unit, the signal can make the character you typed appear on the screen, or it trigger a response based on the command issued by your input.



Fig 4: Types of Keyboard

- a. 101 Key enhanced keyboard: It has one hundred and one keys on it.
- b. 104 Key windows keyboard: It has one hundred and four keys on it.
- c. Ergonomic keyboard: Normal keyboard
- d. Multimedia keyboard: provide media control to play mp3, CD, DVD.
- e. Cordless keyboard: The cordless keyboard is run on battery

Features of a Keyboard

- * Alphanumeric key: a-z, 0-9, punctuation keys, all keys associate with punctuation keys, parenthesis.
- * Virtual keys: (arrow keys, function keys) -
- * Modifier or command keys: [SHIFT, Alternate (ALT), and Control (CTRL)] keys are special keys that modify the action of another key when pressed together.

Each key on a standard keyboard has one or two characters. Press the key to get the lower character and hold Shift to get the upper characters.

Some keyboard keys and their functions

- i. Enter key- to begin the next line, paragraph, or tell the computer to obey a command
- ii. Space bar- to insert spaces between words
- iii. Backspace – to delete characters to the left of the cursor
- iv. Caps lock- to notify the computer when to type upper case and when to type lower case letter
- v. Shift key – to type symbols located on the keyboard with two symbols on then

The mouse: is an input device that is used to send messages (instructions) from the user to the computer system (applications packages). Most modern computers today have two buttons the left one is used to select objects and text and the right one is used to access menus.

Laptop computers use touchpads, buttons, and other devices to control the pointer. Hand-held use a combination of devices to control the pointer, including touch screens.

A mouse performs the following functions

- ☞ A mouse can be used to select icons on the Desktop
- ☞ A mouse can be used to select commands in software applications.
- ☞ It can be used to open a program, a folder, or a file.
- ☞ A mouse can be used to move icons representing programs, folders, or files on the screen.



Fig 5: Examples of Mouse

The most common mouse actions are as follows:

- a) Pointing b) Clicking c) Double-clicking d) Right-clicking
- e) Dragging and dropping f) Highlighting/selecting

Scanner: This is an input device, used to produce electronic copy (soft copy) of a hard copy (text or image) document. Scanners allow you to transfer pictures and photographs to your computer. A scanner 'scans' the image from the top to the bottom, one line at a time, and transfers it to the computer as a sequence of bits or a bitmap. The scanner does not recognize the content of the printed material it is scanning. Everything on the page (text and graphics objects) is converted into one bitmapped graphics image (bitmap), which is a pattern of dots.



Fig. 6: Scanner

Bar Code Reader: The barcode readers are used in places like supermarkets, bookshops, etc. A bar code is a pattern printed in lines of different thickness. The bar-code reader scans the information on the barcodes and transmits it to the computer for further processing. The system gives a fast and error-free entry of information into the computer.



9999999

Fig. 7: Bar Code and Reader

Digital Camera: The digital camera is an input device mainly used to capture images, sound, or video. The digital camera takes a still photograph, stores it, and sends it as a digital input to the computer. It is a modern and popular input device.



Fig. 8: Digital Camera

Light Pen: A light pen is a pointing device shaped like a pen and is connected to a monitor. The tip of the light pen contains a light-sensitive element that, when placed against the screen, detects the light from the screen enabling the computer to identify the location of the pen on the screen.

Fig. 9: Light Pen

Light pens have the advantage of 'drawing' directly onto the screen, but this can become uncomfortable, and they are not accurate.

Touch Sensitive Screen: Touch Sensitive Screen is a type of display screen that has a touch-sensitive panel. It is a pointing device that enables the user to interact with the computer by touching the screen. You can use your fingers to directly touch the objects on the screen and the screen senses the touch and communicate the object selection to the computer.

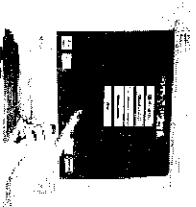


Fig. 10: Touch Sensitive Screen

Magnetic Ink Character Recognition (MICR): MICR is widely used by banks to process cheques. Human readable numbers are printed on documents such as cheque using special magnetic ink. The cheque can be read using a special input unit, which can recognize magnetic ink characters. This method eliminates manual errors. It also saves time, ensures security and accuracy of data.

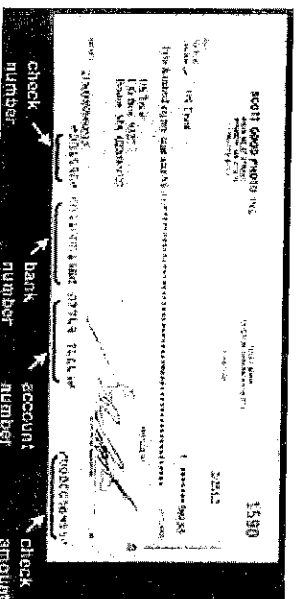


Fig. 11: MICR Cheque

Magnetic Reader: Magnetic reader is an input device that reads a magnetic stripe on a card. It is handy and data can be stored and retrieved. It also provides quick identification of the card's owner.

All the credit cards, ATM cards (banks), petro cards, etc. stores data in a magnetic strip which can be read easily by the magnetic reader

Fig. 12: Magnetic Reader



Optical Character Recognition (OCR): The OCR technique permits the direct reading of any printed character like MICR but no special ink is required. With OCR, a user can scan a page from a book. The computer will recognize the characters on the page as letters and punctuation marks, and stores. This can be edited using a word processor.

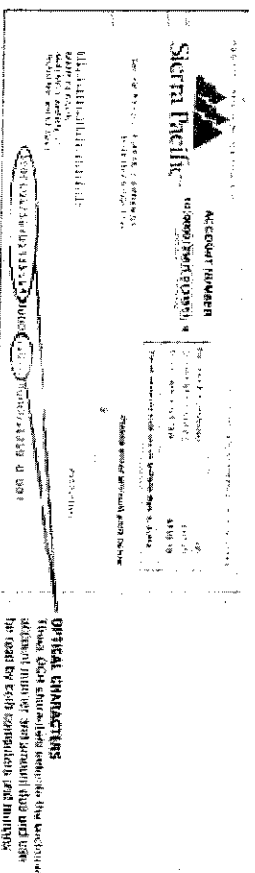


Fig. 13: OCR Sheet

Optical Mark Reading and Recognition (OMR): In this method, special pre-printed forms are designed with boxes that can be marked with a dark pencil or ink. Such documents are read by a reader, which transcribes the marks into electrical pulses that are transmitted to the computer. They are widely used in applications like objective type answer papers evaluation in which a large number of candidates appear, timesheets of factory employees, etc.

Qn. No.	Answers	Ans. No.
1	① ② ● ④	3
2	① ② ③ ●	4
3	① ● ② ④	2
4	① ② ● ④	3
5	● ② ③ -④	1
6	● ② ③ ④	1
7	① ② ③ ●	4

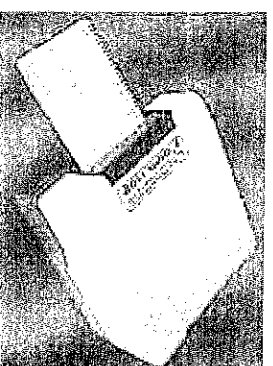


Fig. 14: OMR Reader

Smart Cards: This input device stores data in a microprocessor embedded in the card. This allows information, which can be updated, to be stored on the card. These data can be read and given as input to the computer for further processing. Most of the identification cards use this method to store and retrieve vital information.



Fig.15: Smart Cards

Notes Taker: Notes taker is a device that captures natural handwriting on any surface onto a computer. Using an electronic pen, the **notes taker** displays the user's handwritten notes, memos, or drawings on the computer, and stores the image for future use.

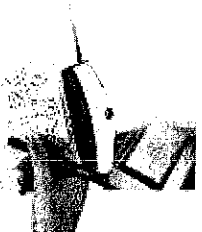


Fig. 16: Notes Taker

Microphone: The microphone serves as a voice input device. It captures the voice data and input to the computer. Using the microphone along with speech recognition software can offer a completely new approach to input information into your computer.

Speech recognition programs, although not yet completely exact, have made great strides in accuracy as well as ease of use. The voice-in or speech recognition approach can almost fully replace the keyboard and mouse. Speech recognition can now open the computer world to those who may have been restricted due to a physical handicap

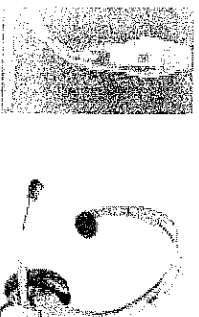


Fig.17: Microphone

Output Devices: Output is anything that comes out of a computer. An output device is capable of presenting information from a computer. There are many output devices attached to computers. But the monitors and printers are commonly used output devices. Disk drives and CD recorders can also be considered output devices because the system sends data to them to be saved

Monitors: The monitor is a commonly used output device, sometimes called a Visual Display Unit (VDU). It provides a visual display of data. Monitors are connected with the computer and are similar in appearance to a television set.

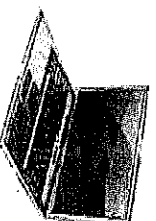


Fig. 18. a) CRT Monitor

b) LCD Monitor

c) Laptop screen

The monitors work like a television screen and also look alike. Most desktop computers use a monitor with a *cathode-ray tube* or *liquid crystal display*. Most notebooks use a liquid crystal display monitor. CRT is a vacuum tube used as a display screen in a video terminal or TV. CRT has higher

radiation, consumes more energy to operate, and it also big compare with LCD. LCD is a flat panel display that uses liquid crystals.

Although almost exclusively used in laptops, LCD technology is also the most popular flat panel desktop monitors. With their low power consumption, low radiation, and space-saving footprint, flat panels are expected to eventually become the standard and gradually replacing the CRT.

The screen resolution can be adjusted by the user to control the screen's display. The smallest dot that can be displayed is called a pixel (picture element). The number of pixels that can be displayed vertically and horizontally gives the resolution of the monitor. The resolution of the monitor determines the quality of the display. Some popular resolutions are 640 x 480 pixels, 800 x 600 pixels and 1024 x 768 pixels. A resolution of 1024 x 768 pixels will produce a sharper image than 640 x 480 pixels

Printers: The printer is an output device that prints text or images on paper or other media. By printing, you create what is known as a 'hard copy'. There are different kinds of printers, which vary in their speed and print quality. The two main types of printers are impact printers and non-impact printers.

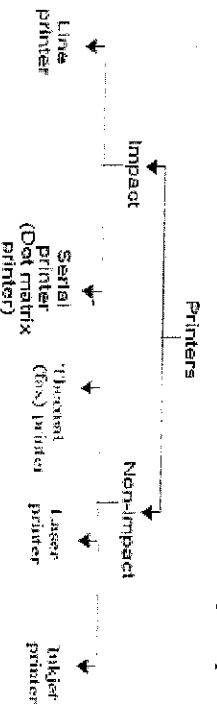


Fig.19 Types of Printers

Impact printers include all printers that print by striking an ink ribbon. Impact printers use a print head containing several metal pins that strike an inked ribbon placed between the print head and the paper. Line printers, dot matrix printers are some of the impact printers.

Characteristics of Impact Printers

1. In impact printers, makes physical contact with the paper to produce an image
2. Due to being robust and low cost, they are useful for bulk printing and printing multiple copies
3. They are mechanical and tend to be slow, very noisy, and do not support transparencies.

Non-impact printers are much quieter than impact printers as their printing heads do not strike the paper. Non-impact printers include laser printers, inkjet printers, and thermal printers.

Characteristics of Non-Impact Printers

1. Non-impact printers are faster than impact printers because they have fewer moving parts.
2. Have the ability to change typefaces automatically but cannot print multipart forms.
3. These printers usually support the transparencies and can produce high-quality graphics

Line Printer: Line printers are high-speed printers capable of printing an entire line at a time. A line printer can print 150 lines to 3000 lines per minute. The limitations of line printer are they can print

only one font, they cannot print graphics, the print quality is low and they are noisy to operate. But it can print a large volume of text data very fast compared to the other printers. It is also used to print on multipart stationaries to prepare copies of a document.

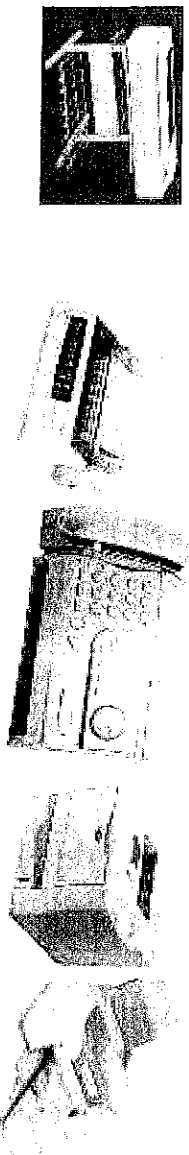


Fig. 20 a) Line Printer b) Dot Matrix c) Thermal Printer d) Laser Printer e) Inkjet Printer

Dot Matrix Printer: The most popular serial printer is the dot matrix printer. It prints one line of 8 or 14 points at a time, with the print head moving across a line. Dot-matrix printers work like a typewriter transferring ink from a ribbon to paper with a sequence or 'matrix' of tiny pins. They are normally slow. The printing speed is around 300 characters per second. It uses multipart stationaries to prepare copies of a document.

Thermal Printer: Thermal printers are printers that produce images by pushing electrically heated pins against special heat-sensitive paper. They are inexpensive and used widely in fax machines and calculators. Thermal printer paper tends to darken over time due to exposure to sunlight and heat. So, the printed matters on the paper fade after a week or two. It also produces a poor-quality print.

Laser Printers: Laser printers use a laser beam and dry powdered ink to produce a fine dot matrix pattern. It can produce a very good quality of graphic images. One of the chief characteristics of laser printers is their resolution – how many dots per inch (dpi) they lay down. The available resolutions range from 300 dpi at the low end to around 1200 dpi at the high end.

Inkjet Printers: Inkjet printers use colour cartridges that combine magenta, yellow and cyan inks to create colour tones. A black cartridge is also used for crisp monochrome output. Inkjet printers work by spraying ionizing ink at a sheet of paper. Magnetized plates in the ink's path direct the ink onto the paper in the described shape.

Speakers: The computer is an electromechanical device that produces voice output (audio data). Speaker serves as a voice output device. Using speakers along with speech synthesizer software, the computer can provide voice output. Voice output has become very common in many places like airlines, banks, automatic telephone inquiry system, etc. Users can also hear music/songs using the



voice output system. Fig.21: Speakers

Plotters: Apart from the output devices like printers, plotters are also used to produce graphical output. Although printer output is very convenient for many purposes, the user needs to present the

information graphically to understand its significance.

Projector: A projector is an optical *output* device that *takes* (projects) images or slides generated by a computer *onto* a projection screen. This surface or a projection screen, may be white screen or sometimes a wall. The advantages of are; Projectors may be used as a cheaper alternative to a television or monitor in large gatherings e.g. in schools, and at home to see the movies etc. The applications of Projectors include:- (a) could be used to demonstrate a presentation on a big screen in a classroom or a conference room for describing a topic, (b) in Companies for meeting and conference purposes etc.

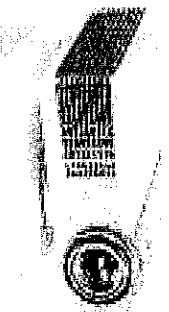


Fig. 22: Projector

Storage Devices

The computer may need to store data, programs, etc. in a computer-readable medium. This is called secondary storage. *Secondary storage* is a type of memory that can hold data for long periods, even when there is no power to the computer. Programs are normally stored in secondary memory and loaded into main memory as needed. Important data, such as word processing documents, payroll data, and inventory records, is saved to secondary storage as well. Secondary Memory is also termed as 'auxiliary' or 'backup' storage, it is typically used as a supplement to main storage. It is much cheaper than the main storage and stores a large amount of data and instructions permanently. Hardware devices like magnetic tapes, Floppy Disk, Hard Disk, Optical disks, Compact Disk (CD), Digital Versatile Disk (DVD), and Blu-ray Disc (BD) fall under this category.

The average time required to reach a storage location and obtain its contents is called its *access time*. In electromechanical devices with moving parts such as disks and tapes, the access time consists of a seek time required to position the read-write head to a location and transfer time required to transfer the data to or from the device.

Hard Disk: Hard disk is a magnetic disk with high speed rotating surfaces coated with a magnetic recording medium, on which you can store computer data. The hard disk is a direct-access storage medium to store and retrieve data randomly. When writing data, a write head magnetizes the particles on the disk surface as either north or south poles. When reading data, a read head converts the magnetic polarizations on the disk surface to a sequence of pulses. The read and write heads are generally combined into a single head unit. There may be more than one read/write head. Data is arranged as a series of concentric rings. Each ring (called a track) is subdivided into several sectors, each sector holding a specific number of data elements (bytes or characters).

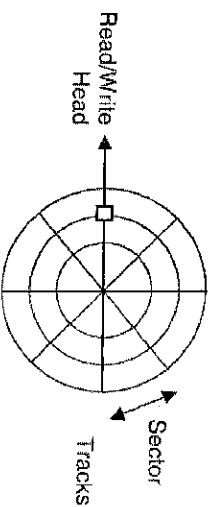


Fig.23: A track subdivided into sectors

The smallest unit that can be written to or read from the disk is a sector. Once a read or write request has been received by the disk unit, there is a delay involved until the required sector reaches the read/write head. This is known as rotational latency, and on average is one half of the period of revolution.

The storage capacity of the disk is determined as **(number of tracks * number of sectors * bytes per sector * number of reading/write heads)**. Thus, the data is stored as magnetized spots arranged in concentric circles (tracks) on the disk. Each track is divided into sectors. The arrangement of tracks and sectors on a disk is known as its 'format'.

High data rates demand that the disk rotates at a high speed (about 3,600 rpm). As the disk rotates read/write heads move to the correct track and fetch the desired data.



Fig. 24: Hard Disk Drive

The storage capacity of a hard disk can be Gigabytes (GB), i.e. thousands of Megabytes of information.

Magnetic Tape: A recording medium consisting of a thin tape with a coating of a fine magnetic strip, used for recording digital data. The tape itself is a strip of plastic coated with a magnetic recording medium.



Fig. 25: Magnetic Tape Reader

Data is stored in frames across the width of the tape. The frames are grouped into blocks or records which are separated from other blocks by gaps. Magnetic tape is a serial access medium, similar to an audio cassette, and so data cannot be randomly located. This characteristic has prompted its use in the regular backing up of hard disks.

Floppy Disk: The floppy drive uses a thin circular disk for data storage. It is a thin, soft magnetic-coated disk contained in a flexible or semi-rigid protective jacket. The disk rotates at 360rpm. A

read/write head makes physical contact with the disk surface. Data is recorded as a series of tracks subdivided into sectors.

The floppy disks are usually 3.5" in size and can hold 1.44 MB of data. Once data is stored on a floppy disk it can be 'write protected' by clicking a tab on the disk. This prevents any new data from being stored or any old data is erased. Disk drives for floppy disks are called floppy drives. Floppy disks are slower to access than hard disks, have less storage capacity, less expensive, and are portable and accessed randomly.



Fig. 26: Floppy Disk

Optical Disk: Optical disks are a storage medium from which data is read and to which it is written by lasers. The optical disk is a random-access storage medium; information can be easily read from any point on the disk. CD-ROM stands for Compact Disk - Read-Only Memory.



Fig. 27: Compact Disk

It is now possible to have CD-ROMs where tracks of information can be written onto them by the user. These are called read/write CD-ROMs and these are becoming a popular and cheap method for storage.

Removable Storage and/or Disk Drives - All disks need the drive to get information read - and put information on the disk - or write. Each drive is designed for a specific type of disk whether it is a CD, DVD, hard disk, or floppy.

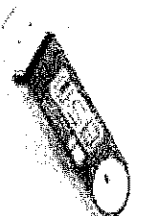


Fig 28: Storage devices

Hard disks store the majority of information on today's modern computer. Information can be stored and deleted as necessary. As files get larger the speed that hard disks can read and write becomes more important.

Digital flash drives work slightly differently as they use memory cards to store information so there are no moving parts. Digital cameras also use Flash memory cards to store information, in this case, photographs. Handheld devices use digital drives and many also use removable or built-in memory cards.